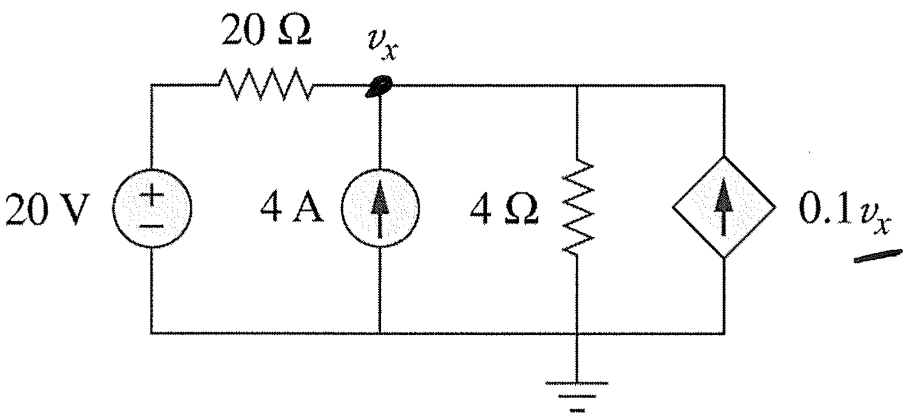
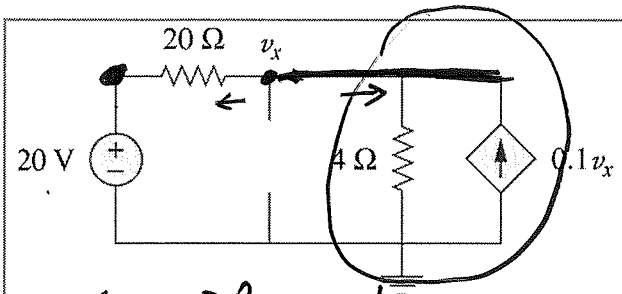
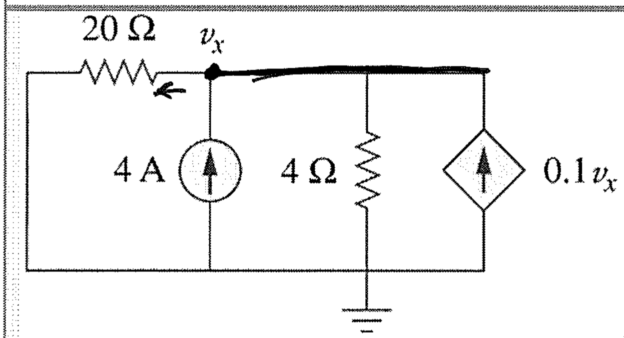
Example #2: Find v_x.

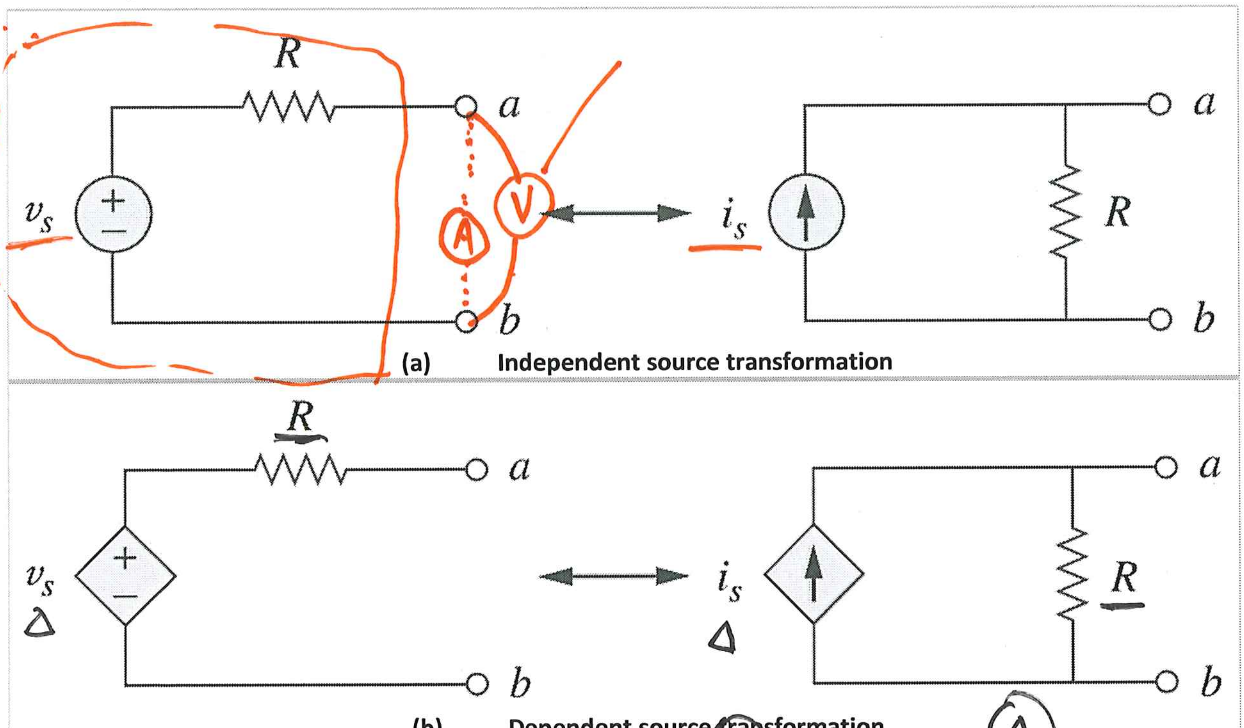
VCCS



 $\frac{V_x - 20}{20} + \frac{V_x}{4} - 0.1V_x = 0$ $V_x - 20 + 5V_x - 2V_x = 0$	$4V_x = 20$ $\Rightarrow V_x^V = 5V$
 $\frac{V_x}{20} + \frac{V_x}{4} = 4 + 0.1V_x$ $V_x + 5V_x = 80 + 2V_x$	$V_x^A = 20V$
Total $V_x = V_x^A + V_x^V = 20V + 5V = 25V$	

L#14: Source transformation

Equivalent circuit: a new circuit whose V-I characteristics are identical with the original circuit.



(a) Independent source transformation

(b) Dependent source transformation

Value:

① change configuration, S → P (V ↑ above S, A ↑ above P)

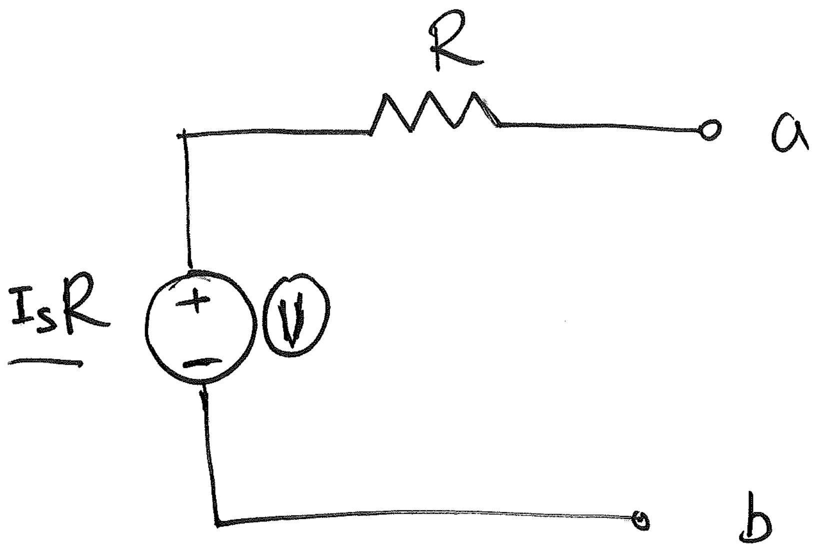
② R: value not change

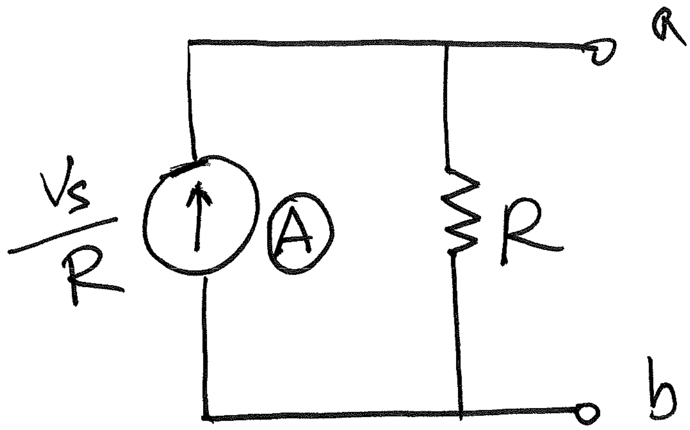
③ V ⇒ A: $I_s = \frac{V_s}{R}$; A ⇒ V: $V_s = I_s R$

Direction:

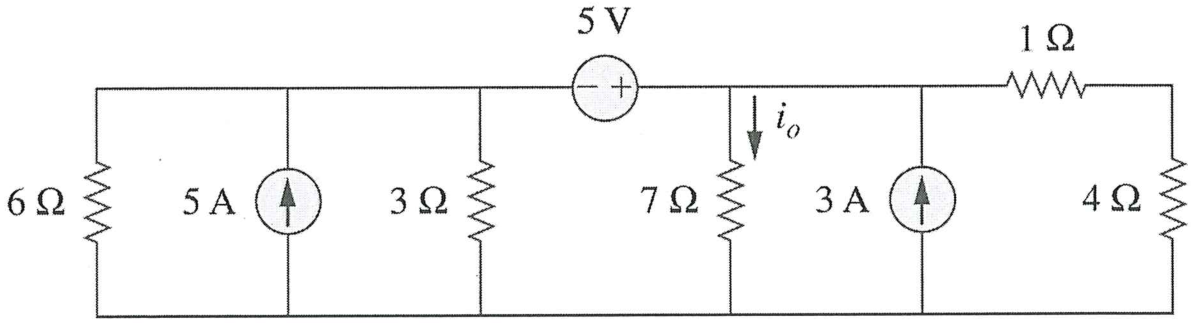
Arrow of A always points to positive end of V

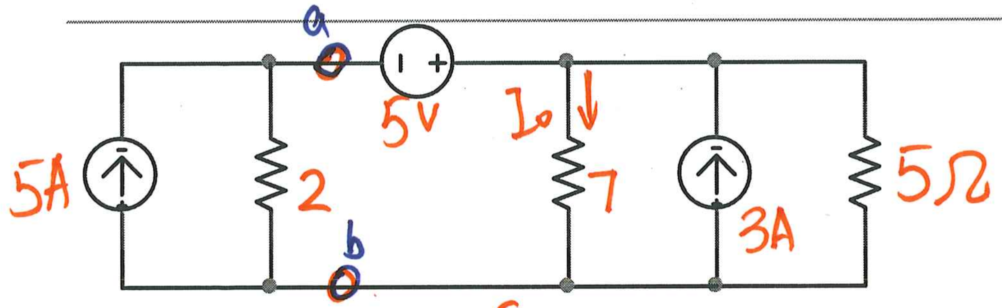
Applicability:


R
a
I_sR
+
−
V
b

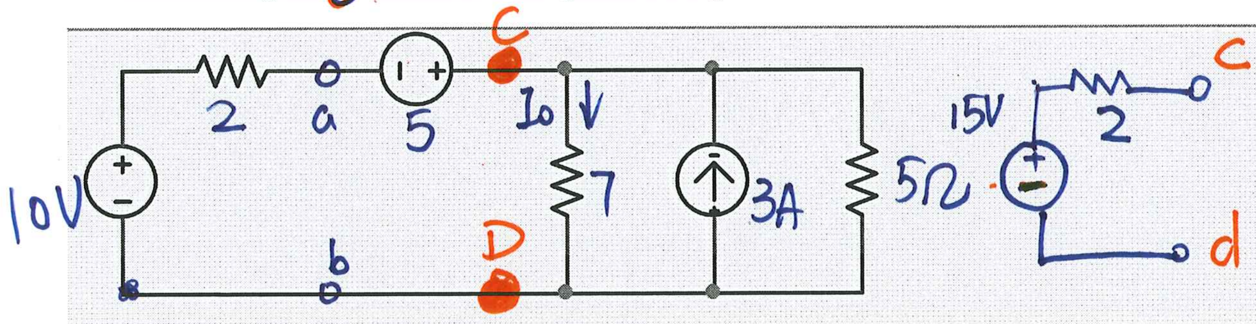

a
$\frac{V_s}{R}$
A
R
b

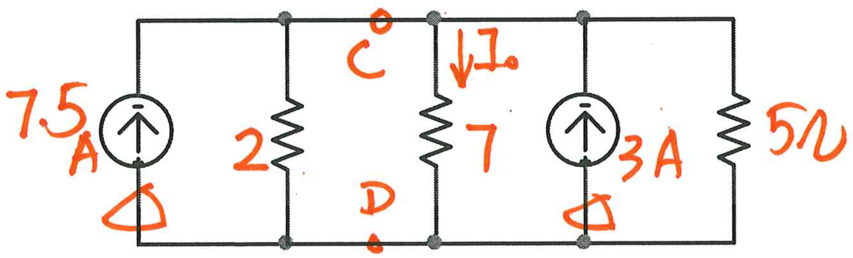
Example #1: Use source transformation and determine i_o.



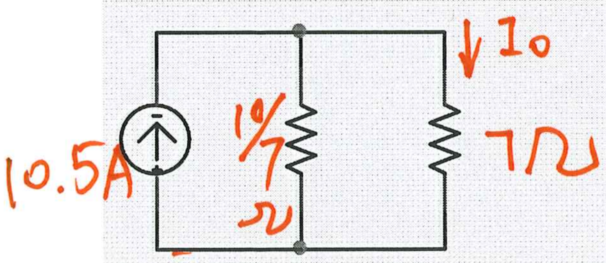


$$\frac{6\times 3}{6+3}=2$$



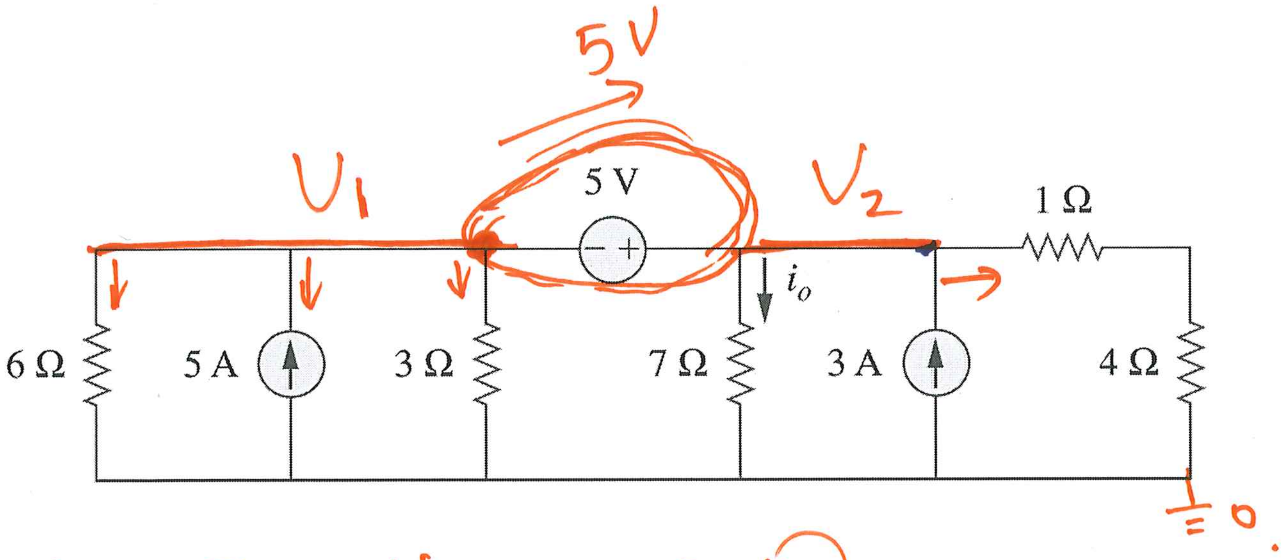


$$\frac{2\times 5}{2+5}$$



$$I_o = 10.5 \cdot \frac{10/7}{7+10/7}$$

$$= \frac{105}{59}\ A$$



$$V_1 + 5 = V_2 \longrightarrow (1)$$

$$\frac{V_1}{6} - 5 + \frac{V_1}{3} + \frac{V_2}{7} - 3 + \frac{V_2}{5} = 0$$

$$\frac{1}{2}V_1 + \frac{V_2}{7} + \frac{V_2}{5} = 8 \quad \xrightarrow{\times 70}$$

$$35V_1 + 24V_2 = 8 \times 70 = 560$$

$$V_1 = V_2 - 5$$

$$35(V_2 - 5) + 24V_2 = 560$$

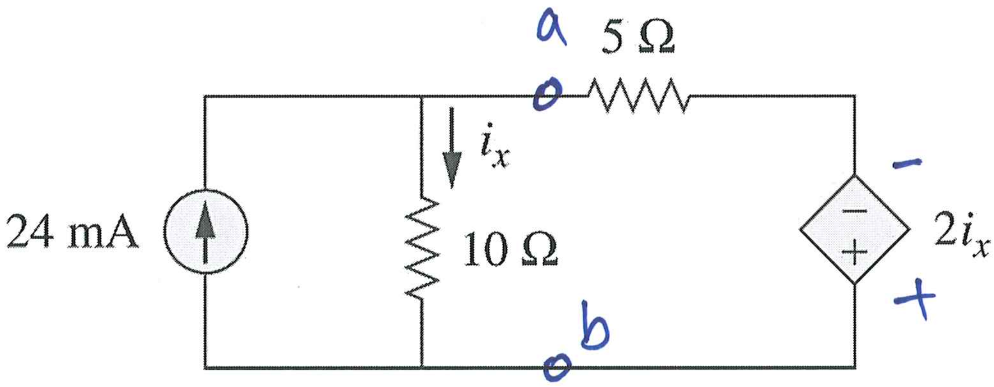
$$59V_2 = 560 + 175$$

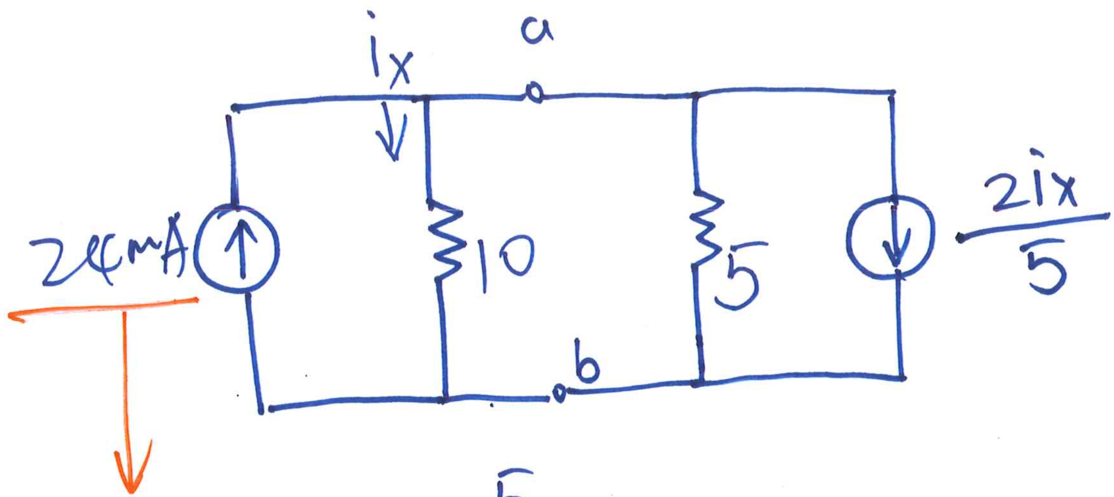
$$\boxed{V_2 = \frac{560+175}{59}}$$

$$I_o = \frac{V_2}{7} = \frac{80+25}{59}$$

$$= \frac{105}{59}$$

Example #2: Use source transformation and determine i_x.





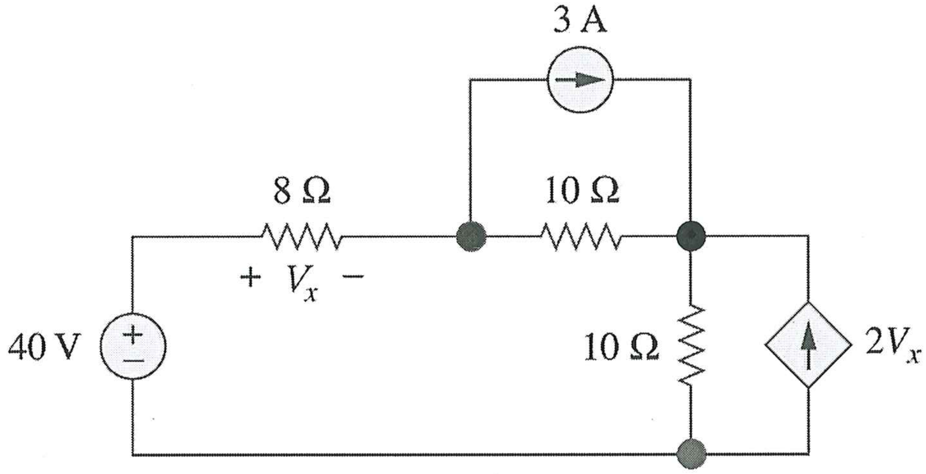
$$\left(24\,\text{mA} - \frac{2i_x}{5}\right) \cdot \frac{5}{10+5} = i_x$$

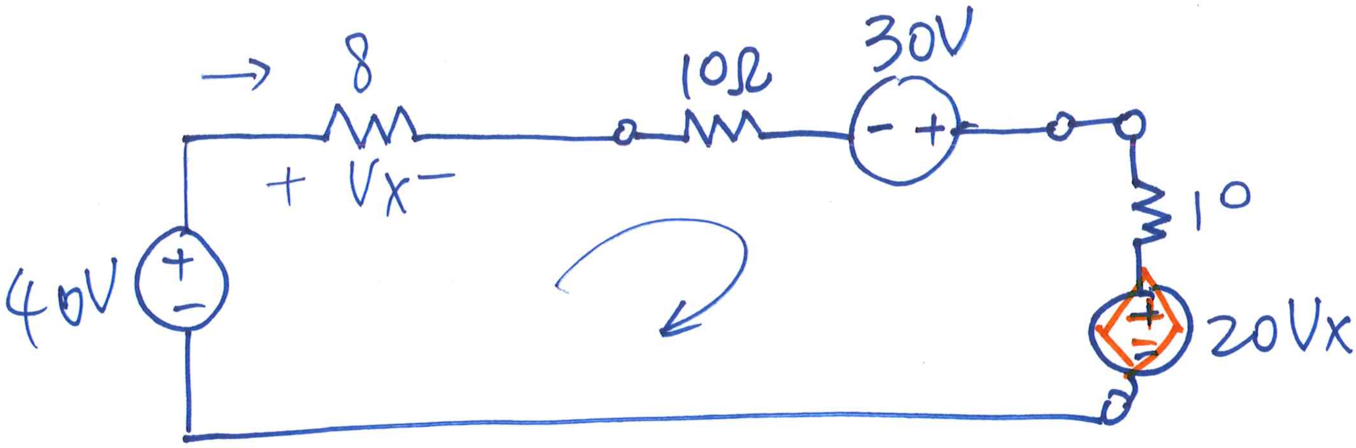
$$24\,\text{mA} - 0.4 i_x = 3 i_x$$

$$\Rightarrow 3.4 i_x = 24\,\text{mA}$$

Do NOT transform the part of the circuit with the unknown variable!

Example #3: Determine V_x.





$$(40+30-20V_x)\cdot\frac{8}{8+10+10}=V_x$$